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Electronic Circuit Design

Lecture notes

Chapter 6: Operational Amplifiers

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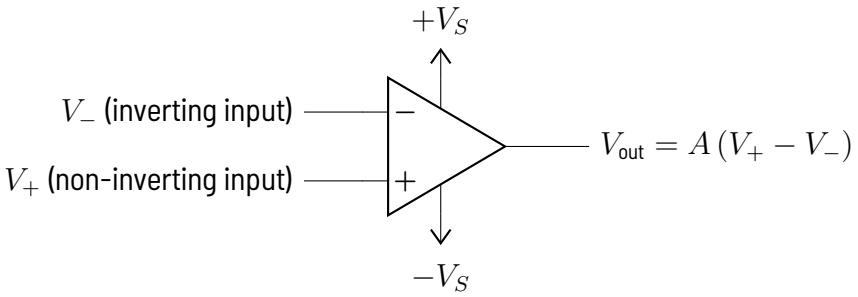
6 Operational Amplifiers

6.1 The Ideal Operational Amplifier

A bridge circuit (chapter 5) turns a small change of a sensor resistance into a voltage of a few millivolts. An analog-to-digital converter needs volts. In between sits the operational amplifier (op-amp): a differential amplifier with a very high gain that is almost never used on its own. Together with a few resistors it becomes a buffer, an amplifier with an exactly defined gain, a summing or a difference amplifier, an instrumentation amplifier or an active filter (chapter 7). The trick behind all of these circuits is negative feedback: the resistors, not the op-amp, define the behaviour.

Chapter 2 introduced the ideal op-amp, the golden rules, the voltage follower and the non-inverting amplifier. This chapter repeats them briefly and derives the other basic circuits with the same two rules. Then we look at the limits of real op-amps: finite open-loop gain, finite bandwidth, output swing, offset voltage and bias currents. At the end we design the measurement amplifier of the lab capstone task "From Sensor to Digital Value". Good introductions are [1] (chapter 4), [5] and [2]; [4] collects many practical details.

The op-amp: symbol and model



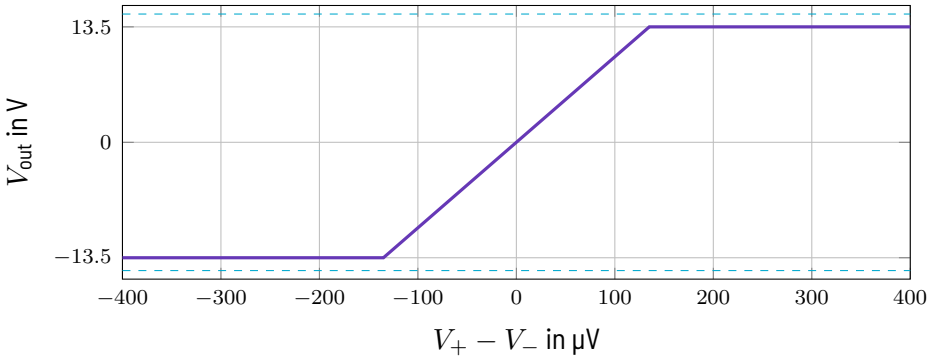
- amplifies the **difference** of its two input voltages
- open-loop gain A: typically $10^5 \ldots 10^7$ (100 dB ... 140 dB)
- supply $\pm V_S$ (or single supply V_S and ground); the output cannot leave the supply range
- supply pins are often omitted in drawings – but they are always there

Properties of the ideal op-amp

property	ideal	real (typical)
open-loop gain A	∞	$10^5 \ldots 10^7$
input resistance	∞	1 MΩ ...1 TΩ
input currents	0	pA (CMOS) ...100 nA (bipolar)
output resistance	0	$10\,\Omega$ ... $100\,\Omega$ (open loop)
offset voltage	0	10 µV ...5 mV
bandwidth	∞	GBW 1 MHz ...100 MHz

The ideal values are good approximations in most circuits of this lecture. Section 6.4 shows when they are not: high gain, high frequency, high source resistance and output voltages close to the supply rails.

Open loop: the op-amp as a comparator



- with $A = 10^5$ and ± 13.5 V output swing the linear range is only ± 135 µV
- any larger difference drives the output to a rail: V_{out} tells only *which input is higher*
- this is a **comparator** – the building block of the flash ADC (chapter 9)

Without feedback an op-amp is useless as a linear amplifier: a difference of a few hundred microvolts is enough to saturate it, and the gain A varies from part to part by a factor of ten and with temperature. As a comparator it works, although dedicated comparator ICs switch much faster and have logic-compatible outputs. The ADC chapter treats comparators in detail.

Negative feedback and the golden rules

- **negative feedback**: a part of V_{out} is fed back to the **inverting** input
- if V_{out} rises, V_- rises, the difference $V_+ - V_-$ shrinks and pulls V_{out} back
- the circuit settles where $V_+ - V_- = V_{\text{out}}/A \approx 0$

Rule 6.1 Golden rules for an op-amp with negative feedback:

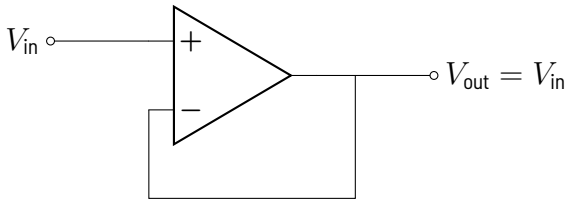
1. The output does whatever is necessary to make the voltage difference between the inputs zero: $V_+ = V_-$.
2. The inputs draw no current: $I_+ = I_- = 0$.

The two rules come from [1]. Rule 1 holds only as long as the feedback loop works, i.e. the output is not saturated at a rail and the feedback goes to the inverting input. With feedback to the non-inverting input (positive feedback) the circuit becomes a Schmitt trigger or an oscillator. Rule 2 is a property of the op-amp itself and holds always (within the bias current).

6.2 Basic Amplifier Circuits

With the golden rules the analysis of the basic circuits takes only a few lines. All circuits in this section exist as Falstad examples. Falstad uses an op-amp with $A = 10^5$ and an output clamped to the supply, which is close enough to ideal for these circuits.

Voltage follower (buffer)



- rule 1: $V_- = V_+$, and $V_- = V_{out} \Rightarrow V_{out} = V_{in}$, gain 1
- input current ≈ 0: the source is **not loaded**
- output: low resistance, can drive mA into a load

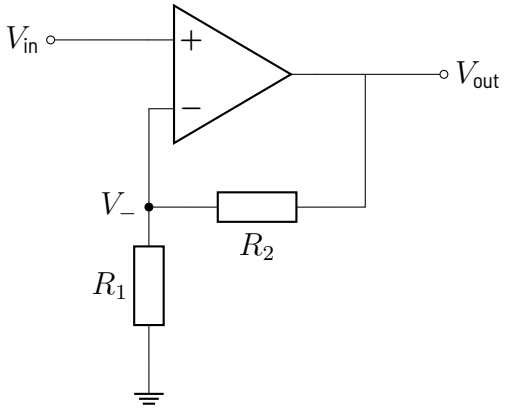
The follower does not amplify the voltage, it amplifies the current: it copies a voltage from a high-impedance node to a low-impedance output. This is exactly what a bridge needs. A bridge tap is a voltage divider with a source resistance of $R_1 \parallel R_2$ (chapter 1); a following stage with a finite input resistance would load it and shift the measured voltage. The same holds for the sample-and-hold circuit (chapter 8): the voltage on the hold capacitor must not be discharged by the following stage, so it is read by a follower with a very small input current.

Why buffering matters

- divider $100\,\text{k}\Omega/100\,\text{k}\Omega$ at $10\,\text{V}$, load $R_L = 10\,\text{k}\Omega$
- directly: $V_{out} = 10\,\text{V} \cdot \frac{R_2 \parallel R_L}{R_1 + R_2 \parallel R_L} = 0.833\,\text{V}$ instead of $5\,\text{V}$
- behind a follower: $5.00\,\text{V}$, the divider does not notice the load
- uses in this course
  - bridge taps → instrumentation amplifier inputs
  - reference ladder of the flash ADC, hold capacitor of the S&H

▶ Falstad: `examples/opamps/opamp_follower.txt`

Non-inverting amplifier

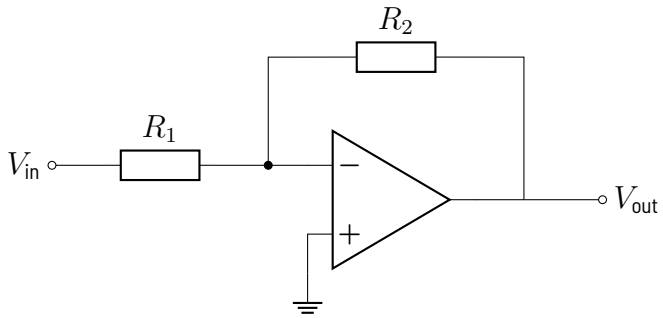


- R_1, R_2 form a divider: $V_- = V_{out} \frac{R_1}{R_1 + R_2}$; rule 1: $V_- = V_{in}$

Non-inverting amplifier

$$G = \frac{V_{out}}{V_{in}} = 1 + \frac{R_2}{R_1} \qquad \text{input resistance} \approx \infty$$

Inverting amplifier

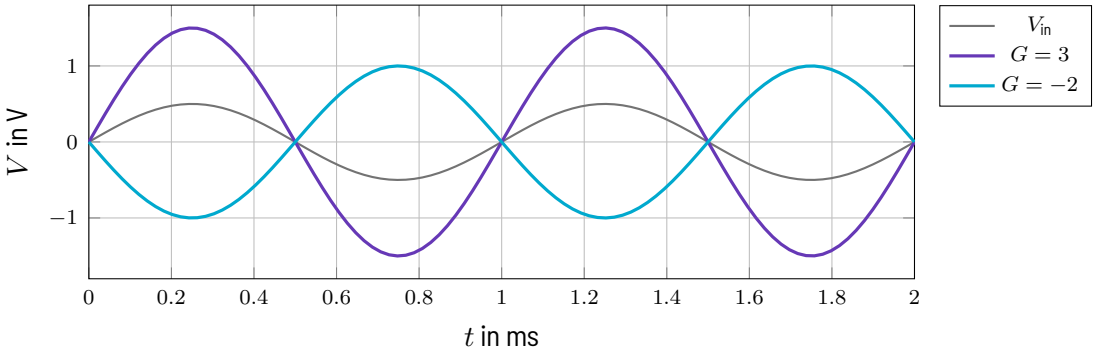


- rule 1: $V_- = V_+ = 0$, the inverting input is a **virtual ground**
- rule 2: the current V_{in}/R_1 continues through R_2: $V_{\text{out}} = -R_2 \cdot V_{\text{in}}/R_1$
- input resistance $= R_1$ (it loads the source!)

Inverting amplifier

$$G = \frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{R_2}{R_1}$$

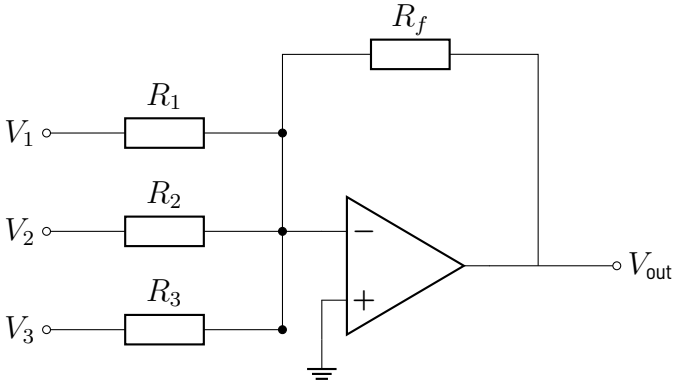
Non-inverting and inverting in Falstad



- V_{in}: 0.5 V amplitude, 1 kHz; $R_1 = 10\,\text{k}\Omega$, $R_2 = 20\,\text{k}\Omega$ in both circuits
- non-inverting: $G = 1 + 2 = 3$, 1.5 V in phase; inverting: $G = -2$, 1.0 V, phase $180°$

▶ Falstad: `examples/opamps/opamp_noninv_inv.txt`

Summing amplifier



- the virtual ground decouples the inputs; the currents add up in R_f

Inverting summing amplifier

$$V_{\text{out}} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$$

Because the summing node is held at 0 V, every input sees only its own resistor and the inputs do not influence each other. Weighted sums are easy: the weight of input k is $-R_f/R_k$. Applications are audio mixers, binary weighted DACs and adding an offset voltage to a signal (level shifting). The Falstad example adds $V_1 = 1$ V through $10\,\text{k}\Omega$, $V_2 = 2$ V through $20\,\text{k}\Omega$ and $V_3 = -0.5$ V through $10\,\text{k}\Omega$ with $R_f = 10\,\text{k}\Omega$ (exercise 6.2).

6.3 Difference and Instrumentation Amplifiers

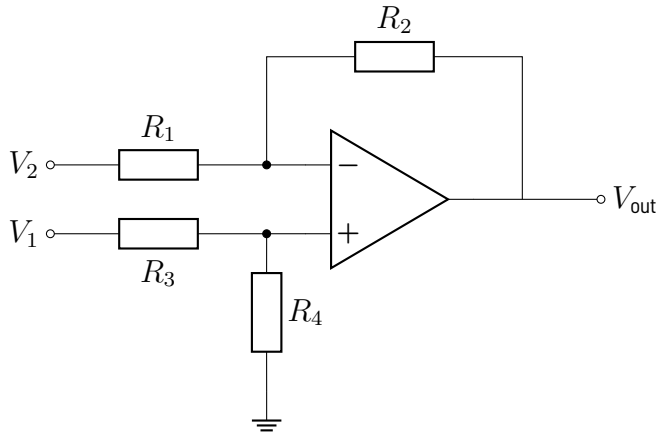
A bridge delivers its signal as the difference of two voltages that both sit at about half the bridge supply. We need an amplifier that amplifies the small difference (the differential-mode signal) and ignores the large common part (the common-mode signal). The difference amplifier does this with one op-amp and four resistors; the instrumentation amplifier adds buffers so that the bridge is not loaded.

Differential and common-mode signal

- two input voltages V_1, V_2 can be written as
  - differential-mode voltage $V_D = V_1 - V_2$ (the measurement signal)
  - common-mode voltage $V_{CM} = (V_1 + V_2)/2$ (not wanted)
- symmetric bridge at $5\,\text{V}$: V_D some mV, $V_{CM} \approx 2.5\,\text{V}$; capstone bridge: $V_D = 0\,\text{mV} \ldots 37\,\text{mV}$, $V_{CM} \approx 0.1\,\text{V}$
- ideal: $V_\text{out} = A_D\, V_D$; real: $V_\text{out} = A_D\, V_D + A_{CM}\, V_{CM}$

Definition 6.1 The **common-mode rejection ratio** is the ratio of differential gain to common-mode gain: $\text{CMRR} = A_D / A_{CM}$, in decibels $20 \log_{10}(A_D/A_{CM})$.

Difference amplifier



- superposition: $V_\text{out} = V_1 \dfrac{R_4}{R_3 + R_4}\left(1 + \dfrac{R_2}{R_1}\right) - V_2 \dfrac{R_2}{R_1}$
- with $R_4/R_3 = R_2/R_1$: $V_\text{out} = \dfrac{R_2}{R_1}(V_1 - V_2)$

The derivation uses superposition. With $V_1 = 0$ the circuit is an inverting amplifier for V_2 (the $+$ input sits at 0 V through R_3 and R_4): $V_\text{out} = -V_2 R_2/R_1$. With $V_2 = 0$ the voltage V_1 is divided by R_3, R_4 and then amplified by a non-inverting amplifier with gain $1 + R_2/R_1$. If the two ratios are equal, the factor in front of V_1 becomes $\frac{R_2/R_1}{1+R_2/R_1}(1 + R_2/R_1) = R_2/R_1$ and only the difference remains. The input resistances are not equal (R_1 for V_2, $R_3 + R_4$ for V_1) and not high: the circuit loads a bridge.

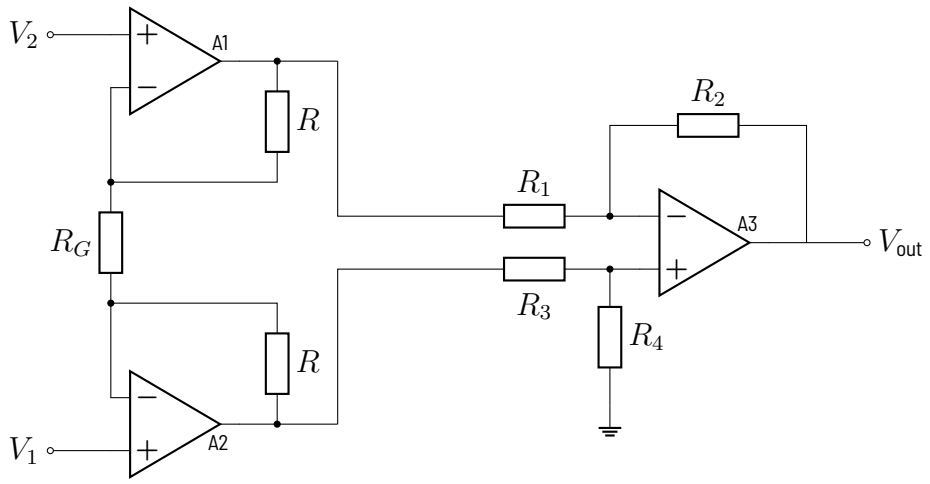
Resistor matching limits the CMRR

- the common-mode rejection depends only on how well R_4/R_3 equals R_2/R_1
- example $G = 10$, $V_1 = 2.505\,\text{V}$, $V_2 = 2.500\,\text{V}$: ideal $V_\text{out} = 50.0\,\text{mV}$
- R_4 only $1\,\%$ too high: $V_\text{out} = 72.6\,\text{mV}$ – an error of $45\,\%$!
- worst case with resistor tolerance ε: $\text{CMRR} \approx \dfrac{1+G}{4\varepsilon}$
  - $1\,\%$ resistors, $G = 10$: $\text{CMRR} \approx 275$ ($49\,\text{dB}$)
  - remedy: $0.1\,\%$ resistors, matched resistor networks, a trimmer, or an integrated difference amplifier (e.g. INA105, AD8276)

▶ Falstad: `examples/opamps/opamp_diffamp.txt`

The common-mode voltage of a symmetric bridge (2.5 V) is more than a hundred times larger than its signal (some 10 mV), so a small common-mode gain already gives a large error. The worst case formula follows if all four resistors deviate by $\pm\varepsilon$ in the unfavourable direction [4]. An op-amp has its own CMRR (typically 80 to 120 dB), but in a discrete difference amplifier the resistors dominate. A higher gain G improves the CMRR, because the differential gain grows while the common-mode error stays the same.

Instrumentation amplifier (three op-amps)



Instrumentation amplifier

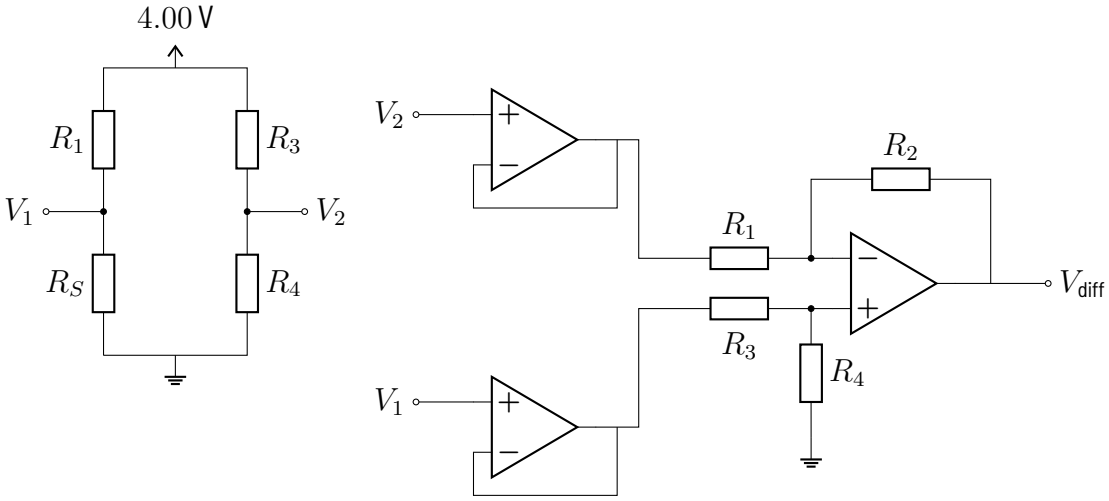
$$V_{\text{out}} = \left(1 + \frac{2R}{R_G}\right) \frac{R_2}{R_1} (V_1 - V_2)$$

How the three op-amp INA works

- A1, A2 are non-inverting: the inputs draw no current, the bridge is not loaded
- rule 1 puts V_1 and V_2 across R_G: current $(V_1 - V_2)/R_G$ flows through R, R_G, R
- difference at the outputs of A1/A2: $(V_1 - V_2)(1 + 2R/R_G)$; the common mode passes with gain 1
- A3 is a difference amplifier with gain R_2/R_1 and removes the common mode
- one resistor R_G sets the gain; integrated INAs (INA128, AD620, AD8221) contain everything except R_G

The input stage amplifies the differential signal by $1 + 2R/R_G$ but the common-mode signal only by one. This improves the CMRR of the complete amplifier by the factor $1 + 2R/R_G$ compared to the difference stage alone. Integrated instrumentation amplifiers have laser-trimmed resistors and reach a CMRR of 100 dB and more. Their reference pin (the lower end of R_4) shifts the output voltage, which is used for level shifting and offset trimming (section 6.5).

The lab version: two buffers and a difference amplifier



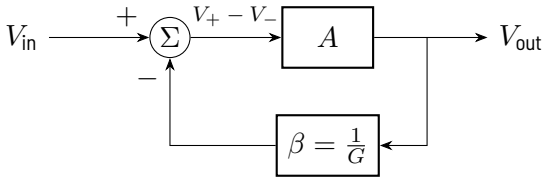
- followers isolate the bridge; the difference stage sets the gain $G_1 = R_2/R_1$
- simpler than the three op-amp version, but all gain must come from R_2/R_1

This is the circuit of the lab: the two buffers make the input resistance practically infinite, so the bridge voltages are not changed by the difference amplifier. Compared to the three op-amp INA the input stage has gain one, so the CMRR is determined by the matching of $R_1 \ldots R_4$ alone. In the capstone circuit the gain is distributed over the difference stage and a following non-inverting stage (section 6.5).

6.4 Limits of Real Op-Amps

The golden rules assume an infinite open-loop gain, zero input currents and an output that can reach any voltage. A measurement amplifier for a bridge works with a total gain of about 100 and an input signal of a few tens of millivolts. Here the deviations of a real op-amp become visible: the gain is a little lower than calculated, the output shows a voltage although the input is zero, and the output cannot reach the supply rails. This section quantifies these effects and shows how to design around them [5, 1].

Finite open-loop gain



- $V_{\text{out}} = A\,(V_{\text{in}} - \beta\,V_{\text{out}})$, with the feedback factor $\beta = 1/G$ (non-inverting: $\beta = R_1/(R_1+R_2)$)
- loop gain $A\beta = A/G$: the larger, the closer the circuit comes to the ideal gain G

Closed-loop gain with finite open-loop gain A

$$G_{\text{real}} = \frac{A}{1+A\beta} = \frac{G}{1+G/A} = \frac{1}{1/G+1/A} \qquad \text{relative error} \approx -\frac{G}{A}$$

How large is the error?

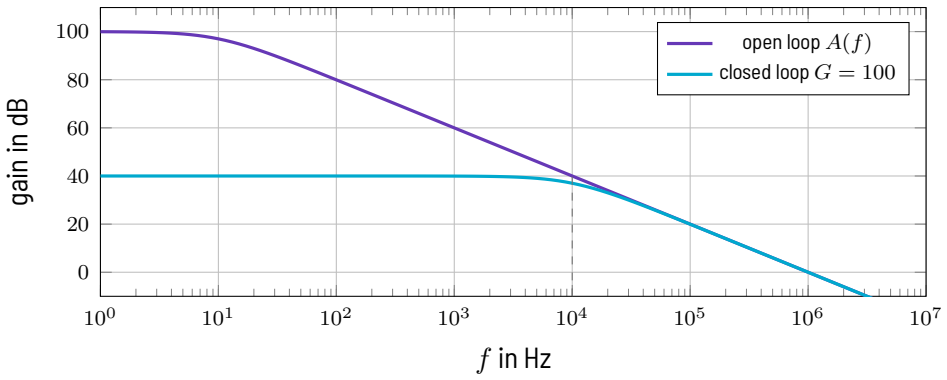
ideal gain G	G_{real} ($A = 10^5$)	error	V_{out} for 1 mV
10	9.999	$-0.01\,\%$	10.0 mV
100	99.90	$-0.1\,\%$	99.9 mV
1000	990.1	$-1\,\%$	0.990 V
10000	9091	$-9.1\,\%$	9.09 V

- the error grows proportionally to G: one stage with $G > 1000$ shows a visible error
- **split the gain**: $G = 10 \cdot 100$ gives 998.9 ($-0.11\,\%$), $G = 100 \cdot 100$ gives 9980 ($-0.2\,\%$)
- rule of the lab: aim for $G < 100$ per stage

▶ Falstad: `examples/opamps/opamp_finite_gain.txt`

The relative errors of cascaded stages add up, but each stage has a small error because its own gain is small: $10/A + 100/A = 110/A$ instead of $1000/A$. Note that A itself is not a precise number: the data sheet gives a minimum, and A decreases with temperature, with the load and, most importantly, with frequency. A circuit that relies on a loop gain of only a few is therefore not precise, even if the formula above corrects for it.

Gain-bandwidth product



- above a low corner frequency (here 10 Hz) A falls by 20 dB per decade: $A(f) \approx \text{GBW}/f$

- closed-loop bandwidth $f_c \approx \text{GBW}/G$ (here 1 MHz/100 = 10 kHz)

Consequences of the GBW

- on the falling slope gain and frequency are inversely proportional:

Gain-bandwidth product
GBW $= A(f) \cdot f = G \cdot f_c$ (constant above the corner frequency)

- at 1 kHz an op-amp with GBW = 1 MHz has only $A = 1000$: a $G = 100$ stage then has $10\,\%$ error
- splitting the gain also helps here: two stages $G = 10$ each have 100 kHz bandwidth
- our sensor signal is slow (a few Hz): the GBW is no problem in the capstone
- **Falstad does not simulate the GBW**: its op-amp has a constant $A = 10^5$ at all frequencies

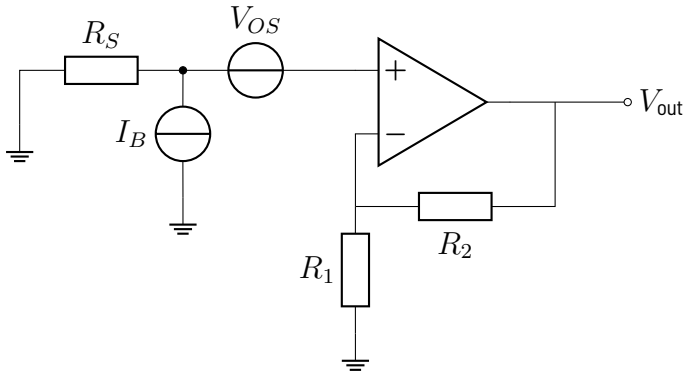
The open-loop gain of almost every op-amp is deliberately rolled off by an internal capacitor (dominant-pole compensation), so that the amplifier stays stable with any amount of negative feedback. The result is a first-order low-pass with a very low corner frequency. The product of gain and frequency is constant on the falling slope and is listed in the data sheet as gain-bandwidth product (GBW) or unity-gain frequency f_T. For the closed loop the gain that counts is the noise gain $1/\beta$, which is $1 + R_2/R_1$ for both the non-inverting and the inverting amplifier. A related limit for large signals is the slew rate, the maximum rate of change of the output voltage (e.g. $0.5\,\text{V}/\mu\text{s}$). In Falstad and in our simulator the op-amp has no bandwidth limit, so results of fast circuits must be checked against the data sheet.

Output swing and rail-to-rail

- the output cannot exceed the supply; a classic op-amp stays 1 V … 2 V away from the rails
  - LM358 at 5 V: output from about 0 V to 3.5 V
- **rail-to-rail output**: reaches the rails within some 10 mV (light load) to a few 100 mV (mA load)
- the **input common-mode range** is limited as well: not every op-amp works with inputs near the rails (**rail-to-rail input**); the capstone bridge sits at only 0.1 V
- consequence for the capstone: an output of exactly 0 V is not possible on a single 5 V supply – use a negative supply, or a small offset and a range of e.g. 0.1 V … 4.1 V

When the output hits a rail the feedback loop is broken, rule 1 no longer holds, and the signal is clipped. Clipping cannot be repaired by later stages: a clipped 50 Hz interference becomes a DC error after the low-pass filter. Each stage therefore needs headroom for the signal plus all interference it has to carry. Falstad's op-amp model clamps exactly at the values `vmax` and `vmin`, which behaves like an ideal rail-to-rail output.

Input offset voltage and bias currents



- offset V_{OS}: a small voltage source in series with an input, amplified by $1 + R_2/R_1$
- bias current I_B flows through the source resistance R_S and adds $R_S I_B$
- $V_{\text{out}} = (1 + R_2/R_1)\,(V_{OS} + R_S I_B)$ with $V_{\text{in}} = 0$

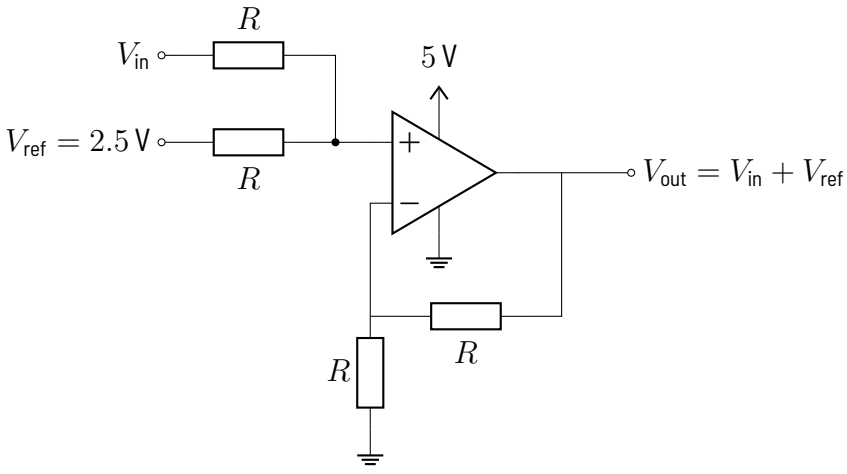
Offset after a high gain

- typical values
  - general purpose bipolar (LM358): $V_{OS} \approx 2$ mV, $I_B \approx 20$ nA
  - CMOS (MCP6002, TLV9002): $V_{OS} \approx 1$ mV, $I_B \approx 1$ pA
  - zero-drift / chopper (OPA2333, MCP6V02): $V_{OS} < 10$ µV
- Falstad example with $G = 101$: $V_{OS} = 1$ mV gives 0.101 V; $I_B = 100$ nA through 100 kΩ gives -1.01 V
- capstone: 2 mV offset $\times$ 107.6 = 0.22 V – 5 % of the full scale, almost half an LSB!
- remedies: zero-drift op-amps, CMOS inputs, low source resistances, **offset trimming**

▶ Falstad: `examples/opamps/opamp_offset.txt`

The offset voltage is a property of the input transistor pair: the two transistors are never perfectly equal. It is specified as a typical and a maximum value and it drifts with temperature (typically 1 to 10 µV/K). Because it appears at the input, it is amplified by the full noise gain. A bridge signal of 37 mV and an offset of 2 mV make an error of more than 5 %. Bias currents are the base currents (bipolar) or gate leakage currents (CMOS) of the input transistors. In a bridge amplifier the source resistance is low (about 100 Ω for the capstone bridge), so I_B matters little; but behind a sample-and-hold capacitor or a high-ohmic filter it does. The Falstad model of the op-amp has no offset and no bias current; the example models them with an extra voltage source and current source.

Single-supply operation and level shifting



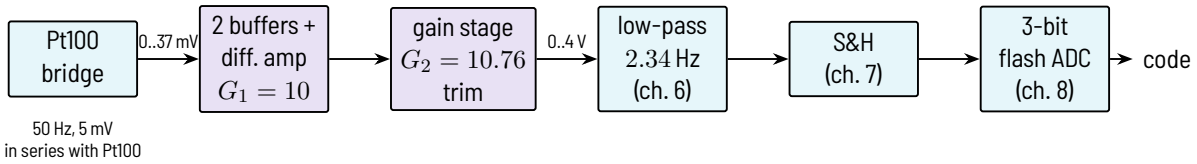
- single supply: the output can only be positive; bipolar signals need a **bias point** (often $V_S/2$)
- here $V_+ = (V_{\text{in}} + V_{\text{ref}})/2$, gain 2: $V_{\text{in}} = \pm 2$ V becomes 0.5 V … 4.5 V
- a follower on the same supply clips the negative half wave at 0 V

▶ Falstad: `examples/opamps/opamp_level_shift.txt`

6.5 Designing the Measurement Amplifier

Now we design the front end of the capstone task (chapter 10). The sensor is a Pt100 ($100\,\Omega$ at $0\,^{\circ}$C, $138.5\,\Omega$ at $100\,^{\circ}$C) in a Wheatstone bridge (chapter 5). The bridge is fed from the same 4.00 V reference that later supplies the ADC ladder, with $R_1 = R_3 = 3.9\,\text{k}\Omega$ and $R_4 = 100\,\Omega$. It is balanced at $0\,^{\circ}$C and delivers 0 to 37.18 mV over the measurement range. The instrumentation amplifier amplifies this to 0 V at the bottom and exactly 4.00 V at the top of the range. The following low-pass filter (chapter 7), the sample-and-hold (chapter 8) and the 3-bit flash ADC (chapter 9) expect exactly this range: with 8 codes over 4 V one LSB is 0.5 V.

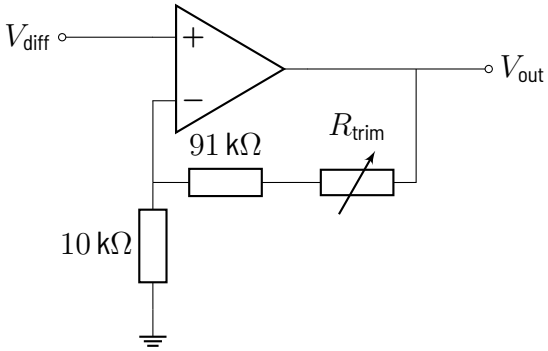
The signal chain of the capstone



- bridge from $4.00\,\text{V}$: $V_{BR} = V_{\text{ref}} \left(\frac{R}{R_1+R} - \frac{R_4}{R_3+R_4} \right) = 0\,\text{mV} \ldots 37.18\,\text{mV}$ for $R = 100\,\Omega \ldots 138.5\,\Omega$
- total gain $G = 4.00\,\text{V}/37.18\,\text{mV} = 107.6$
- split: difference amplifier $G_1 = 100\,\text{k}\Omega/10\,\text{k}\Omega = 10$, non-inverting stage $G_2 = 10.76$ – both < 100
- zero: set by the bridge ($R_4 = R(0\,°\text{C})$); span: one resistor R_F in the second stage

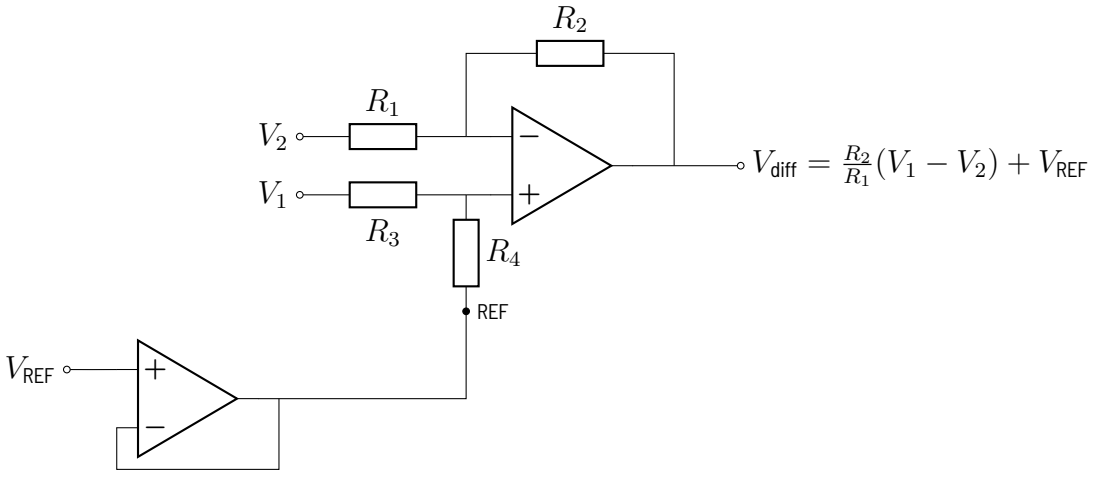
The common-mode voltage of this bridge is only about $0.1\,\text{V}$ (R_4 is small compared with R_3), so the op-amps must accept input voltages close to their negative supply. With a symmetric supply or an op-amp whose input range includes ground this is no problem.

Gain stage with trimmer



- required: $G_2 = 4.00\,\text{V}/(10 \cdot 37.18\,\text{mV}) = 10.758 \Rightarrow R_F = 97.58\,\text{k}\Omega$
- fixed E96 value: $R_F = 97.6\,\text{k}\Omega$, $G_2 = 10.76$, full scale $4.0005\,\text{V}$ (used in Falstad)
- in hardware: $R_F = 91\,\text{k}\Omega + 10\,\text{k}\Omega$ trimmer: $G_2 = 10.1 \ldots 11.1$, full scale $3.76\,\text{V} \ldots 4.13\,\text{V}$; setting $R_{\text{trim}} = 6.6\,\text{k}\Omega$
- the fixed resistor carries most of the value: fine trimmer resolution, enough range for the tolerances

Zero adjustment at the reference input



- the lower end of R_4 is the **reference input**: V_{REF} adds directly to the output
- it must be driven from a low impedance (follower), otherwise it changes R_4 and the CMRR
- in the capstone the zero is set by the bridge ($R_4 = R(0\,°\text{C})$); a REF adjustment of a few mV is needed only if op-amp offsets or resistor tolerances shift it

Trimming procedure

- two adjustments: zero (offset) first, then span (gain)

Design card: Measurement amplifier (zero and span)

1. Calculate the bridge span $V_{BR,\text{max}}$ and the total gain $G = V_{\text{FS}}/V_{BR,\text{max}}$ (here $4.00\,\text{V}/37.18\,\text{mV} = 107.6$).
2. Split the gain: input stage (buffers + difference amplifier) $G_1 = 10$, second stage $G_2 = G/G_1 = 10.76$, each stage $G < 100$.
3. Use a fixed resistor for most of R_F and a trimmer for the rest ($91\,\text{k}\Omega$ + $10\,\text{k}\Omega$).
4. **Zero**: set the sensor to the bottom of the range ($100\,\Omega$, or a precise resistor) and check $V_{\text{out}} = 0.00\,\text{V}$; correct with R_4 or V_{REF} if necessary.
5. **Span**: set the sensor to the top of the range ($138.5\,\Omega$), adjust R_{trim} until $V_{\text{out}} = 4.00\,\text{V}$.
6. Check the zero again.

The order matters: the span trimmer changes the gain of everything that arrives at the second stage, including a remaining offset. If the zero is correct first, a later change of the gain no longer moves the zero point. In the capstone the zero is fixed by the bridge and the span by R_F, so the two adjustments do not influence each other.

The complete front end in Falstad

- bridge $4.00\,\text{V}$, $R_1 = R_3 = 3.9\,\text{k}\Omega$, $R_4 = 100\,\Omega$, Pt100 with a slider from $100\,\Omega$ to $138.5\,\Omega$
- two followers, difference amplifier $10\,\text{k}\Omega/100\,\text{k}\Omega$, gain stage $1+(91\,\text{k}\Omega+R_{\text{trim}})/10\,\text{k}\Omega$, $R_{\text{trim}} = 6.6\,\text{k}\Omega$
- readings: $V_R = 0.100\,\text{V}$, $V_{BR} = 37.18\,\text{mV}$, $V_{\text{diff}} = 372\,\text{mV}$, $V_{\text{out}} = 4.00\,\text{V}$ at $138.5\,\Omega$; $V_{\text{out}} = 0.00\,\text{V}$ at $100\,\Omega$
- the 50 Hz interference of $5\,\text{mV}$ in series with the Pt100 is a *differential* signal: $4.9\,\text{mV}$ at the tap, $\times 107.6 = 0.52\,\text{V}$ amplitude
  - no CMRR helps here – the low-pass filter of chapter 7 has to remove it
  - the output must have headroom: $4\,\text{V} + 0.52\,\text{V}$ peak

▶ Falstad: `examples/opamps/opamp_ina.txt`

6.6 Exercises

Exercise 6.1: Non-inverting and inverting amplifier

A sine voltage with $0.5\,\text{V}$ amplitude and $1\,\text{kHz}$ is to be amplified. Resistors: $R_1 = 10\,\text{k}\Omega$, $R_2 = 20\,\text{k}\Omega$.

a) Calculate gain and output amplitude of the non-inverting amplifier.

b) Calculate gain and output amplitude of the inverting amplifier. What is the phase shift?

c) What input resistance does the source see in both circuits?

d) Which resistor would you change to get an inverting gain of -4.7?

e) Check a) and b) in Falstad.

Solution 6.1

a) $G = 1 + R_2/R_1 = 1 + 20/10 = 3$; output amplitude $3 \cdot 0.5\,\text{V} = 1.5\,\text{V}$, in phase with the input.

b) $G = -R_2/R_1 = -2$; output amplitude $1.0\,\text{V}$, phase shift $180°$.

c) Non-inverting: practically infinite (the op-amp input, MΩ to TΩ). Inverting: $R_1 = 10\,\text{k}\Omega$, because the inverting input is a virtual ground.

d) $R_2 = 4.7 \cdot R_1 = 47\,\text{k}\Omega$ (E12 value).

e) ▶ Falstad: `examples/opamps/opamp_noninv_inv.txt`
the scope shows `Vout_ni` with $\pm 1.50\,\text{V}$ and `Vout_inv` with $\pm 1.00\,\text{V}$, inverted.

Exercise 6.2: Summing amplifier

An inverting summing amplifier has $R_f = 10\,\text{k}\Omega$ and three inputs: $V_1 = 1\,\text{V}$ via $R_1 = 10\,\text{k}\Omega$, $V_2 = 2\,\text{V}$ via $R_2 = 20\,\text{k}\Omega$, $V_3 = -0.5\,\text{V}$ via $R_3 = 10\,\text{k}\Omega$.

a) Calculate V_{out} and the voltage at the inverting input.

b) Calculate the current through R_f.

c) How must R_3 be changed so that V_3 is weighted with -0.25?

d) Check a) in Falstad.

Solution 6.2

a) $V_{\text{out}} = -10\,\text{k}\Omega\left(\frac{1\,\text{V}}{10\,\text{k}\Omega} + \frac{2\,\text{V}}{20\,\text{k}\Omega} + \frac{-0.5\,\text{V}}{10\,\text{k}\Omega}\right) = -(1 + 1 - 0.5)\,\text{V} = -1.50\,\text{V}$. The inverting input is a virtual ground: $V_- = 0\,\text{V}$.

b) $I = 0.1\,\text{mA} + 0.1\,\text{mA} - 0.05\,\text{mA} = 0.15\,\text{mA}$, which gives $V_{\text{out}} = -10\,\text{k}\Omega \cdot 0.15\,\text{mA} = -1.5\,\text{V}$.

c) Weight $-R_f/R_3 = -0.25 \Rightarrow R_3 = 40\,\text{k}\Omega$ (e.g. $39\,\text{k}\Omega$ + $1\,\text{k}\Omega$).

d) ▶ Falstad: `examples/opamps/opamp_summing.txt`
`Vout` $= -1.50\,\text{V}$, `V_sum_node` $= 0.00\,\text{V}$.

Exercise 6.3: Difference amplifier and resistor matching

A difference amplifier with $R_1 = R_3 = 10\,\text{k}\Omega$ and $R_2 = R_4 = 100\,\text{k}\Omega$ amplifies a bridge signal: $V_1 = 2.505\,\text{V}$, $V_2 = 2.500\,\text{V}$.

a) Calculate differential and common-mode voltage and the output voltage.

b) R_4 is $1\,\%$ too high ($101\,\text{k}\Omega$). Calculate the output voltage.

c) Calculate the common-mode gain and the CMRR for b).

d) Estimate the worst-case CMRR with $1\,\%$ resistors.

e) Check a) and b) in Falstad.

Solution 6.3

a) $V_D = 5\,\text{mV}$, $V_{CM} = 2.5025\,\text{V}$; $V_{\text{out}} = 10 \cdot 5\,\text{mV} = 50.0\,\text{mV}$.

b) $V_+ = 2.505\,\text{V} \cdot \frac{101}{111} = 2.279\,32\,\text{V}$, $V_{\text{out}} = V_+\,(1+10) - 2.5\,\text{V} \cdot 10 = 25.0726\,\text{V} - 25\,\text{V} = 72.6\,\text{mV}$: $45\,\%$ too high.

c) $A_{CM} = 11 \cdot \frac{101}{111} - 10 = 0.0090$ (input both at $2.5\,\text{V}$ gives $22.5\,\text{mV}$); $A_D \approx 10$; CMRR $= 10/0.009 \approx 1100$ ($61\,\text{dB}$).

d) CMRR $\approx (1+G)/(4\varepsilon) = 11/0.04 = 275$ ($49\,\text{dB}$): with $2.5\,\text{V}$ common mode up to $9\,\text{mV}$ error at the input, i.e. almost half of the full bridge span. Matched resistors or an integrated difference amplifier are needed.

e) ▶ Falstad: `examples/opamps/opamp_diffamp.txt`
`Vout_matched` $= 50.0\,\text{mV}$, `Vout_mismatch` $= 72.6\,\text{mV}$.

Exercise 6.4: Finite open-loop gain

A signal of $1\,\text{mV}$ DC is amplified with non-inverting stages ($R_1 = 1\,\text{k}\Omega$). The op-amps have $A = 10^5$ (as in Falstad).

a) Calculate V_{out} for one stage with $G = 1000$ and one stage with $G = 10000$. Give R_2 and the errors.

b) Calculate V_{out} for $G = 10 \cdot 100$ and $G = 100 \cdot 100$ in two stages.

c) How large is the error of the $G = 100$ stage at $1\,\text{kHz}$ if the op-amp has GBW $= 1\,\text{MHz}$?

d) Check a) and b) in Falstad.

Solution 6.4

a) $G = 1000$: $R_2 = 999\,\text{k}\Omega$, $G_{\text{real}} = 1000/(1+0.01) = 990.1$, $V_{\text{out}} = 0.990\,\text{V}$ ($-1\,\%$). $G = 10000$: $R_2 = 9.999\,\text{M}\Omega$, $G_{\text{real}} = 10000/1.1 = 9091$, $V_{\text{out}} = 9.09\,\text{V}$ ($-9.1\,\%$).

b) $10/(1+10^{-4}) \cdot 100/(1+10^{-3}) = 9.999 \cdot 99.90 = 998.9$: $V_{\text{out}} = 0.999\,\text{V}$ ($-0.11\,\%$). $(100/1.001)^2 = 9980$: $V_{\text{out}} = 9.98\,\text{V}$ ($-0.2\,\%$).

c) $A(1\,\text{kHz}) = 1\,\text{MHz}/1\,\text{kHz} = 1000$, error $\approx G/A = 10\,\%$ (and a phase shift). Falstad does not show this, because it does not simulate the GBW.

d) ▶ Falstad: `examples/opamps/opamp_finite_gain.txt`
`Vout_1000` $= 0.990\,\text{V}$, `Vout_10x100` $= 0.999\,\text{V}$,
`Vout_10000` $= 9.09\,\text{V}$, `Vout_100x100` $= 9.98\,\text{V}$.

Exercise 6.5: Offset voltage and bias current

A non-inverting amplifier has $R_1 = 1\,\text{k}\Omega$, $R_2 = 100\,\text{k}\Omega$; its input is at $0\,\text{V}$.

a) The op-amp has an offset voltage of $1\,\text{mV}$. Calculate V_{out}.

b) The input is connected to ground through $R_S = 100\,\text{k}\Omega$ and a bias current of $100\,\text{nA}$ flows into the non-inverting input (no offset). Calculate V_{out}.

c) The capstone amplifier has $G = 107.6$ and the op-amps have 2 mV offset (referred to the input). Compare the output error with the full scale of 4 V and with one LSB of the 3-bit ADC.

d) Check a) and b) in Falstad.

Solution 6.5

a) $V_{\text{out}} = (1 + 100) \cdot 1\,\text{mV} = 0.101\,\text{V}$.

b) $V_+ = -R_S I_B = -100\,\text{k}\Omega \cdot 100\,\text{nA} = -10\,\text{mV}$, $V_{\text{out}} = 101 \cdot (-10\,\text{mV}) = -1.01\,\text{V}$. A CMOS op-amp with $I_B = 1$ pA would give only 10 µV.

c) $107.6 \cdot 2\,\text{mV} = 0.215\,\text{V}$: 5.4 % of the full scale, 0.43 LSB (LSB $= 4\,\text{V}/8 = 0.5\,\text{V}$). The zero must be checked and trimmed, or a zero-drift op-amp used.

d) ▶ Falstad: `examples/opamps/opamp_offset.txt`
`Vout_offset` $= 0.101$ V, `V_plus_b` $= -10.0$ mV, `Vout_bias` $= -1.01$ V.

Exercise 6.6: Capstone: instrumentation amplifier with 4.00 V full scale

The Pt100 of the capstone ($100\,\Omega$ at 0 °C, $138.5\,\Omega$ at 100 °C) sits in a bridge fed from the 4.00 V reference: R_1 (3.9 kΩ) above the Pt100 gives V_S, $R_3 = 3.9\,\text{k}\Omega$ above $R_4 = 100\,\Omega$ gives V_R. The amplifier consists of two followers, a difference amplifier and a non-inverting stage. Its output shall be 0 V at 0 °C and exactly 4.00 V at 100 °C.

a) Calculate V_R, V_S and the bridge output $V_{BR} = V_S - V_R$ at 0 °C and 100 °C. What is the common-mode voltage?

b) Calculate the total gain. Why is it split into two stages?

c) Choose $G_1 = 10$ for the difference amplifier and give its resistors.

d) Dimension the second stage with $R_g = 10\,\text{k}\Omega$. Give the exact R_F, the nearest E96 value and a version with 91 kΩ plus a 10 kΩ trimmer (setting and full-scale range).

e) The three op-amp INA could replace the followers. Which R_G gives an input stage gain of 10 with $R = 10\,\text{k}\Omega$?

f) A 50 Hz interference of 5 mV amplitude is in series with the Pt100. How large is it at the output? What follows for the supply voltage and the next stage?

g) Check the design in Falstad.

Solution 6.6

a) $V_R = 4\,\text{V} \cdot 100/4000 = 0.100\,\text{V}$. 0 °C: $V_S = 0.100\,\text{V}$, $V_{BR} = 0$ (sensor current 1.000 mA). 100 °C: $V_S = 4\,\text{V} \cdot 138.5/4038.5 = 0.137\,18\,\text{V}$, $V_{BR} = 37.18\,\text{mV}$. The common mode is only about 0.1 V: the op-amp inputs must work close to the negative rail.

b) $G = 4.00\,\text{V}/37.18\,\text{mV} = 107.6$. One stage would be possible (error $G/A = 0.11\,\%$), but two stages of about 10 keep each error at 0.01 % and the bandwidth high, follow the lab rule $G < 100$, and separate zero (bridge) and span (R_F).

c) $R_1 = R_3 = 10\,\text{k}\Omega$, $R_2 = R_4 = 100\,\text{k}\Omega$ (0.1 % or matched). $V_{\text{diff}} = 10 \cdot 37.18\,\text{mV} = 371.8\,\text{mV}$ at 100 °C.

d) $G_2 = 4.00\,\text{V}/371.8\,\text{mV} = 10.758$, $R_F = 9.758 \cdot 10\,\text{k}\Omega = 97.58\,\text{k}\Omega$. E96: 97.6 kΩ, $G_2 = 10.76$, full scale 4.0005 V. Trimmer version: 91 kΩ + $R_{\text{trim}} = 6.6\,\text{k}\Omega$; range $G_2 = 10.1 \ldots 11.1$, full scale 3.76 V … 4.13 V.

e) $1 + 2R/R_G = 10 \Rightarrow R_G = 2R/9 = 2.22\,\text{k}\Omega$; then the difference stage can have gain 1 and the second stage stays as it is.

f) At the tap: $5\,\text{mV} \cdot \frac{3.9\,\text{k}\Omega}{3.9\,\text{k}\Omega + 100\,\Omega} = 4.875\,\text{mV}$, at the output $\times 107.6 = 0.52\,\text{V}$ amplitude: the output swings up to $4.52\,\text{V}$. The op-amps need headroom (e.g. $\pm 15\,\text{V}$ as in Falstad); the low-pass of chapter 7 must attenuate the interference below $\text{LSB}/2 = 0.25\,\text{V}$.

g) ▶ Falstad: `examples/opamps/opamp_ina.txt`
with the Pt100 at $138.5\,\Omega$ the labels show `VB_minus` $= 0.100\,\text{V}$, `V_diff` $= 372\,\text{mV}$ and `Vout` $= 4.00\,\text{V}$; with the slider at $100\,\Omega$ `Vout` $= 0.00\,\text{V}$.

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